Chung-En (Johnny) **Yu**

PH.D. CANDIDATE · GRADUATE RESEARCH ASSISTANT · AI RESEARCHER

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"Towards Reliable Multimodal LLM-based Agentic AI Systems with Metacognitive Awareness"

Summary

Chung-En (Johnny) Yu is a Ph.D. candidate in Intelligent Systems and Robotics at the University of West Florida and the Institute for Human Machine Cognition, where he works as a Graduate Research Assistant in the Artificial Superintelligence (ASI) Lab under Prof. Brian Jalaian, with research mentorship and sponsorship from Prof./LTC Nathaniel D. Bastian of the United States Military Academy (USMA), West Point since 2024. His research focuses on reliable agentic AI systems, multimodal foundation models, uncertainty quantification, and neurosymbolic AI for high-stakes real-world applications such as cybersecurity and remote sensing, with peer-reviewed publications and manuscripts.

Research Interests Include But Are Not Limited To:

- **Major:** Reliable Agentic AI Systems, Multimodal Foundation Models, Uncertainty Quantification, Adversarial and Hallucination Robustness
- **Minor:** Neurosymbolic AI, Robust Control Systems, Robotics and SLAM, Computer Vision

Education

University of West Florida & Institute for Human Machine Cognition — *Pensacola, FL, USA*
PH.D. IN INTELLIGENT SYSTEMS AND ROBOTICS — *2023 – Expected 2027*

University of Minnesota — *Saint Paul, MN, USA*
M.S. IN ELECTRICAL AND COMPUTER ENGINEERING — *2021 – 2023*

National Taipei University — *New Taipei City, Taiwan*
B.S. IN ELECTRICAL ENGINEERING — *2016 – 2020*

Work Experience

University of West Florida & Institute for Human Machine Cognition — *Pensacola, FL, USA*
GRADUATE RESEARCH ASSISTANT, ARTIFICIAL SUPERINTELLIGENCE LAB — *Aug. 2023 – Present*

- Advised by Prof. Brian Jalaian and Prof./LTC Nathaniel D Bastian.
- Conducted research on reliable multimodal LLM-based agentic AI systems, with a focus on uncertainty quantification, hallucination detection, prediction abstention, iterative self-critique, cross-model verification, and multi-agent debate.
- Developed training-free agentic and uncertainty-aware frameworks for vision-language reasoning and remote sensing tasks, improving robustness and accuracy while maintaining efficient inference-time overhead.
- Applied neurosymbolic AI and post-hoc uncertainty modeling to cybersecurity problems, including open-set network intrusion detection and classification reliability under challenging settings.

Industrial Technology Research Institute — *Hsinchu, Taiwan*
SUMMER INTERN, AUTONOMOUS MOBILE ROBOT DEVELOPMENT TEAM — *Jul. 2022 – Aug. 2022*

- Advised by Dr. Andy Jeng.
- Developed a computationally efficient, real-time collision avoidance algorithm for autonomous delivery robots using low-cost LiDAR sensors.
- Implemented the solution in ROS with C++, processing point cloud data to identify obstacles and navigable space on hardware without GPU acceleration.

National Taipei University — *New Taipei City, Taiwan*
UNDERGRADUATE RESEARCH ASSISTANT, INTELLIGENT CONTROL SYSTEM DIGITAL IMAGE PROCESSING LAB — *Aug. 2018 – Jun. 2019*

- Advised by Prof. Hooman Samani.
- Designed and implemented an industrial-level robotic arm system using computer vision for precise object detection and calibration.
- Developed a user interface in TwinCAT and coordinated robot motion control via C++ and PLC for industrial automation workflows.

Publications

Chung-En Yu is listed as Yu, C.-E. Underlining denotes the first-author or co-first-author contribution.

Conference & Workshop Papers

- Yu, C.-E., Jalaian, B., & Bastian, N. D. (2026). **SCoOP: Semantic Consistent Opinion Pooling for Uncertainty Quantification in Multiple Vision Language Model Systems**. Presented at *ICLR 2026 Workshop on Agentic AI in the Wild: From Hallucinations to Reliable Autonomy*.
- Yu, C.-E., Jalaian, B., & Bastian, N. D. (2026). **Visual Reasoning Agent: Robust Vision Systems in Remote Sensing via Inference-Time Scaling**. In *MORS 2026 Artificial Intelligence Workshop Proceedings*.
- Yu, C.-E., Jalaian, B., & Bastian, N. D. (2026). **ORCA: An Agentic Reasoning Framework for Hallucination and Adversarial Robustness in Vision-Language Models**. In *Proceedings of ACM ICCBDC 2026*.
- Yu, C.-E., Jalaian, B., & Bastian, N. D. (2024). **Mitigating Large Vision-Language Model Hallucination at Post-hoc via Multi-agent System**. In *Proceedings of the AAAI Symposium Series*, Vol. 4, No. 1, pp. 110–113. Presented at *AAAI 2024 Fall Symposium*.
- Sander, J., Yu, C.-E., Jalaian, B., & Bastian, N. D. (2024). **Uncertainty-Quantified Neurosymbolic AI for Open Set Recognition in Network Intrusion Detection**. In *MILCOM 2024 – IEEE Military Communications Conference*, pp. 13–18. IEEE.
- Bizzarri, A., Yu, C.-E., Jalaian, B., Riguzzi, F., & Bastian, N. D. (2024). **Neuro-Symbolic Integration for Open Set Recognition in Network Intrusion Detection**. In *International Conference of the Italian Association for Artificial Intelligence*, pp. 50–63. Springer.

Journal

- Bizzarri, A., Yu, C.-E., Jalaian, B., Riguzzi, F., & Bastian, N. D. (2025). **Neurosymbolic AI for network intrusion detection systems: A survey**. *Journal of Information Security and Applications*, Vol. 94, Article 104205.

Submitted Manuscripts & Preprints

- Yu, C.-E., Jalaian, B., & Bastian, N. D. (2025). **Hydra: An Agentic Reasoning Approach for Enhancing Adversarial Robustness and Mitigating Hallucinations in Vision-Language Models**. *arXiv preprint arXiv:2504.14395*.
- Bizzarri, A., Yu, C.-E., Jalaian, B., Riguzzi, F., & Bastian, N. D. (2024). **A Synergistic Approach in Network Intrusion Detection by Neurosymbolic AI**. *arXiv preprint arXiv:2406.00938*.

Selected Projects

Applied ML and Robotics Projects

- **ML/DL Kaggle Competitions** (Python, 2024): Created applied ML/DL projects for paraphrase detection with Transformers, satellite image classification using CNNs, and structured data modeling with logistic regression, decision trees, KNN, and random forests.
- **Bayesian Inference on Adelie Penguin Flipper Length** (R, 2024): Applied a Normal-Normal conjugate Bayesian model, computed posterior distributions and credible intervals, and conducted Bayes Factor analysis.
- **Deep Reinforcement Learning for Atari** (Python, PyTorch, 2022): Engineered a Double Deep Q-Network from scratch and trained it on classical Atari environments.
- **Cooperative Air-Ground Robot Localization** (MATLAB, 2022): Developed a Kalman filter-based optimal estimation system for UAV-assisted ground robot localization under simulated GPS-denied conditions.
- **SLAM for Autonomous Navigation Robot** (Python, ROS, Webot, 2022): Implemented an Extended Kalman Filter within ROS for autonomous exploration and localization in unknown, bounded environments.
- **Temporal Logic Path Planning for Persistent Surveillance** (MATLAB, 2021): Developed formally verified path-planning algorithms for cooperative UAV-UGV surveillance teams.
- **Edge Computing for Facial Recognition Monitoring** (Microsoft Azure IoT Edge, 2020): Deployed a real-time facial recognition system using Raspberry Pi and Azure IoT Edge with low computational overhead.

Teaching Experience

University of West Florida *Pensacola, FL, USA*

CO-INSTRUCTOR, TEACHING ASSISTANT, GRADUATE-LEVEL ML/DL COURSE *Jan. 2024 – Present*

- Co-designed assignments, projects, lecture slides, live coding demonstrations, and research spotlights for graduate-level machine learning and deep learning courses.
- Mentored students through classical machine learning and deep learning workflows, including model development, evaluation, and applied coding exercises.

Academic Service

USMA-Hosted Technical Exchange Meetings

INVITED PRESENTER *2024 – 2026*

- Invited four times to present in semi-annual technical exchange meetings hosted by the United States Military Academy (USMA), contributing to AI research discussions with academic and defense-oriented stakeholders.

MORS AI Workshop; IEEE MILCOM; IEEE TGRS; ACM TAISAP

AD HOC REVIEWER *2024 – 2026*

- Peer-reviewed submissions for Military Operations Research Society (MORS) AI Workshop, IEEE Military Communications Conference (MILCOM), IEEE Transactions on Geoscience and Remote Sensing (TGRS), and ACM Transactions on AI Security and Privacy (TAISAP).

AAAI 2024 Fall Symposium (ATRACC Track)

WEBSITE CHAIR *Jun. 2024 – Nov. 2024*

- Designed, developed, and maintained the official website for the AI Trustworthiness and Risk Assessment for Challenged Contexts (ATRACC) track, coordinating content, schedule, submission, and registration updates with organizers.

Honors & Awards

2023	**Four-Year Graduate Assistantship**, Full tuition waiver and stipend from University of West Florida for the Ph.D. program	*Pensacola, FL, USA*
2020	**Venture Capital Award**, Innovation & Entrepreneurship Competition from Ministry of Education; won $100K NTD among 300+ teams	*Taiwan*
2019	**Honorable Mention**, Data Science Application Hackathon at National Taipei University; the only honorable mention among 20 teams	*New Taipei City, Taiwan*
2018	**Vice-Chairperson**, Department Student Association at National Taipei University; managed a budget of over $170K NTD and coordinated large-scale cross-department initiatives	*Taiwan*

Skills

- **Programming:** Python, PyTorch, C, C++, MATLAB, R.
- **AI agents & tools:** Claude Code, Codex, HuggingFace, LangChain, LangGraph, Ollama, vLLM, AutoGen.
- **Machine learning & data science:** LLMs, VLMs, DRL, CNNs, NSAI, DNNs, RNNs, Decision Trees, Ensemble Learning, Regression, Clustering, Bayesian Analysis.
- **Systems & tools:** Linux, Docker, CUDA, High-performance Computing (HPC), SLURM, Git, W&B.
- **Robotics:** ROS, Gazebo, Webot, SLAM, Optimal Estimation, Kalman Filtering, Robust Control.
- **Languages:** Mandarin (Native), English (Proficiency).